

Instruction:

For the homework, a dataset will be provided to you on Canvas, and you need to download it for use. A final Jupyter Notebook file (.ipynb) needs to be submitted as your answers to the homework questions.

You are strongly encouraged to work on this problem set together in the form of a study group. However, everyone should submit his/her individual write-up.

Questions:

The Monetary Policy Committee (MPC) in the Bank of England is responsible for making decisions about Bank Rate. Usually, it's composed of nine members, including the governor, the chief economist and others.

The data provided is from minutes from MPC meetings, which is titled "mpc_minutes.txt". Please use this data to answer the following questions. The total points are 100.

1. (20 points) Load the MPC data, and pre-process the data.
 - lower-case text
 - expand contractions
 - split the documents into tokens
 - remove non-ascii characters, remove short tokens, remove punctuation and numbers, remove stopwords
 - stemming and lemmatization
2. (15 points) Count and the frequency.
 - generate document frequency (df) ranking for stems
 - show the top 10 stems
 - plot the df ranking
3. (25 points) Term frequency inverse document frequency (TF-IDF).
 - generate TF-IDF ranking of stems
 - show the top 10 stems
 - plot TF-IDF ranking
 - remove stems with a TF-IDF score $\leq 3,500$
 - apply the removal and create a new document-term matrix, show the matrix (num_documents x num_terms)
4. (40 points) LDA topic model.
 - create a LDA object with 30 topics (set the hyperparameter of the Dirichlet prior on topics to 200/number of unique vocabulary elements, set the hyperparameter of the Dirichlet prior on document-topic distributions to 50/number of topics, set the number of sampling iterations be 5000, set the random_state to make sure

re-productivity)

- estimate the model (may take 5-10 minutes)
- show the distribution of topics over documents and probability distribution over the vocabulary
- show the top 15 words with highest probability per topic
- create a figure for the first 15 topics
- choose three topics and interpret them (try to summarize the likely topics)